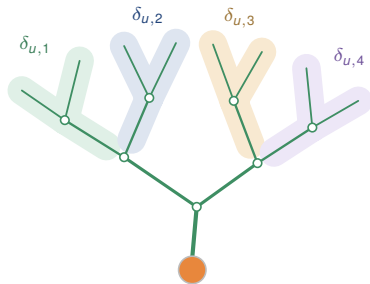

When dendritic structure helps local credit assignment

Coordinates, subtree addresses, route gain, and the alignment boundary

Houman Safaai · Maceo Richards · Bernardo L. Sabatini
Kempner Institute, Harvard University · Harvard Medical School

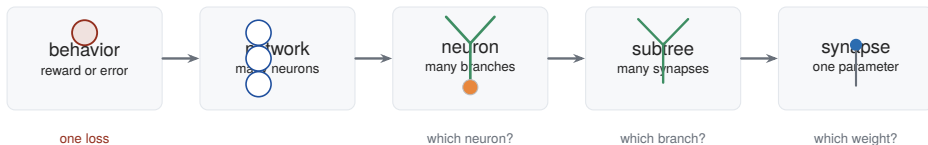
Kempner Learning Dynamics Workshop, 2026



one neuron, multiple candidate credit routes
task error arrives at the soma

Learning is a credit-assignment problem

A global consequence must ultimately guide local synaptic change.



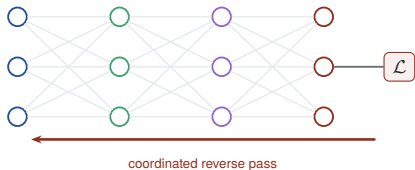
THE ASSIGNMENT

which parameter changes? which direction? how much?

The loss is global, but every update must be local and correctly addressed.

Backpropagation provides exact parameter credit

It is the optimization reference against which local rules are compared.



EXACT UPDATE

$$\Delta W_{ji}^{\ell} = -\eta \underbrace{h_i^{\ell-1}}_{\text{presynaptic activity}} \underbrace{\delta_j^{\ell}}_{\text{parameter-specific error}}$$

REVERSE RECURSION

$$\delta^{\ell} = (W^{\ell+1})^{\top} \delta^{\ell+1} \odot f'(\mathbf{a}^{\ell})$$

Every weight receives a signed magnitude and an exact destination.

Backpropagation answers credit assignment exactly, one parameter at a time.

Literal backpropagation is not a neuronal mechanism

The question is which information a synapse can actually access.

- WEIGHT TRANSPORT

reuse exact downstream weights
in reverse

- NONLOCAL STATE

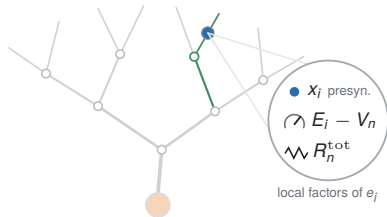
downstream derivatives at the
same learning event

- GLOBAL COORDINATION

ordered Jacobian products
across layers

- PARAMETER INDEXING

a distinct error coordinate for
every destination



what a synapse can measure locally

Biological theories replace, approximate, infer, or physically instantiate different pieces of this reverse computation.

A plausible rule must say what replaces the exact synaptic error.

Local-learning theories differ in what replaces the exact error

Different mechanisms often resolve credit only to the level of a neuron.

- MODULATORY SCALAR

$$\Delta w_i \propto e_i m$$

three-factor rules

- FEEDBACK ALIGNMENT

$$\hat{\delta} = B\delta^{\text{out}}$$

fixed/random routes

- STATE INFERENCE

$$\hat{\delta} \sim s^+ - s^-$$

predictive/equilibrium

- ERROR COMPARTMENTS

$$\hat{\delta}_u \text{ beside } a_u$$

apical/basal models

one scalar

one coordinate

inferred coordinate

few compartments

Frémaux & Gerstner 2016; Lillicrap et al. 2016; Scellier & Bengio 2017; Whittington & Bogacz 2017; Guerguiev et al. 2017; Sacramento et al. 2018; Payeur et al. 2021.

The unresolved spatial step begins after a neuron-level teaching coordinate arrives.

Point-neuron learning stops at one neuronal coordinate

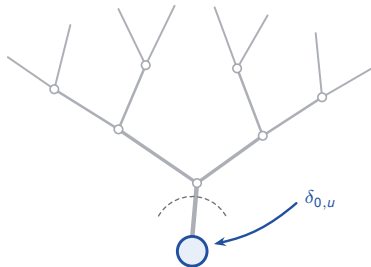
point neuron: $s_u = \sum_i w_i x_i$, $V_u = f(s_u)$

POINT-NEURON RULE

$$\frac{\partial \mathcal{L}}{\partial w_i} = \underbrace{x_i f'(s_u)}_{\text{synapse-local eligibility}} \underbrace{\delta_u}_{\text{one shared coordinate}}$$

Different inputs keep weights and activities distinct, but no variable can assign different teaching coordinates to two subtrees of the same neuron.

The bottleneck is feedback bandwidth, not a collapse of all neurons into one.



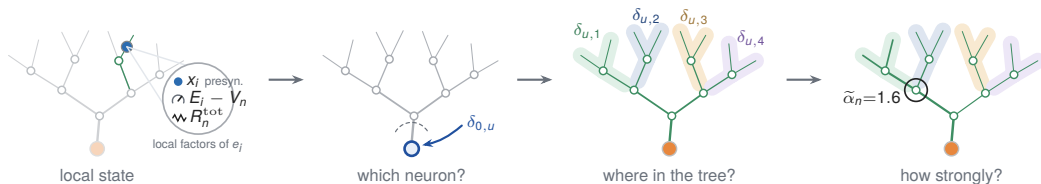
one coordinate reaches the soma

Point neurons distinguish synapses through eligibility, but cannot explicitly address locations inside the arbor.

A dendritic tree adds state, addresses, and route gain

PREVIEW

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{e_i}_{\text{local state}} \underbrace{\delta_{0,u}}_{\text{neuron coordinate}} \underbrace{\tilde{\alpha}_{n(i)}}_{\text{transport/path gain}}$$



The tree adds spatial structure after the neuronal coordinate arrives; whether it helps is an alignment question.

A dendritic neuron is a conductance tree

The steady state is a conductance-weighted quotient at every compartment.

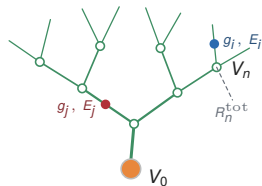
STEADY-STATE COMPARTMENT

$$V_n = R_n^{\text{tot}} \left(\sum_{i \in \mathcal{S}_n} g_i x_i E_i + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c + g_n^{\text{sh}} E_I + g_n^L E_L \right)$$

$$R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}, \quad g_n^{\text{tot}} = g_n^L + \sum_i g_i x_i + \sum_c g_{c \rightarrow n}^{\text{den}} + g_n^{\text{sh}}$$

- SHUNTING

A local inhibitory conductance raises the denominator, alters the compartment state, and edits the transport operator.



signals integrate from leaves to soma



a local shunt edits descendant paths

Shunting is central because it changes both the forward state and the backward sensitivity operator.

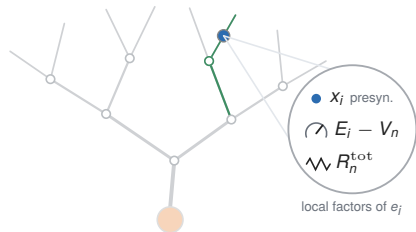
The conductance quotient yields an exact local eligibility

QUOTIENT RULE

$$\frac{\partial V_n}{\partial g_i} = \underbrace{x_i}_{\text{presynaptic activity}} \underbrace{R_n^{\text{tot}}}_{\text{input resistance}} \underbrace{(E_i - V_n)}_{\text{driving force}} \equiv e_i$$

The same synapse has access to all three factors: its presynaptic activity, local voltage, and local conductance state.

In an additive-current model the denominator does not depend on the synaptic weight; the driving-force term is absent.



all factors are available at the synapse

Conductance normalization preserves locality and creates the exact driving-force factor.

Each compartment receives a transported error field

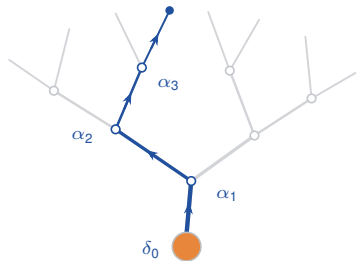
On a tree, every compartment has one unique route to the soma.

COMPARTMENT ERROR

$$\delta_n \equiv \frac{\partial \mathcal{L}}{\partial V_n} = \delta_{0,u} \tilde{\alpha}_n$$

PATH GAIN

$$\tilde{\alpha}_n = \prod_{(j \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_j(V_j) R_k^{\text{tot}} g_{j \rightarrow k}^{\text{den}}$$



the somatic error transported to one site

The route gain is state-dependent and multiplicative; a shunt changes factors on descendant paths.

The teaching signal at a synapse is a somatic coordinate multiplied by transport along its branch path.

Exact dendritic credit factorizes at every synapse

The factorization separates what is local from what must be routed.

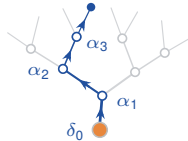
THEOREM

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_{n(i)}^{\text{tot}} (E_i - V_{n(i)})}_{\text{local eligibility } e_i} \underbrace{\delta_{0,u} \tilde{\alpha}_{n(i)}}_{\text{transported error } \delta_{n(i)}}$$



computed locally

×



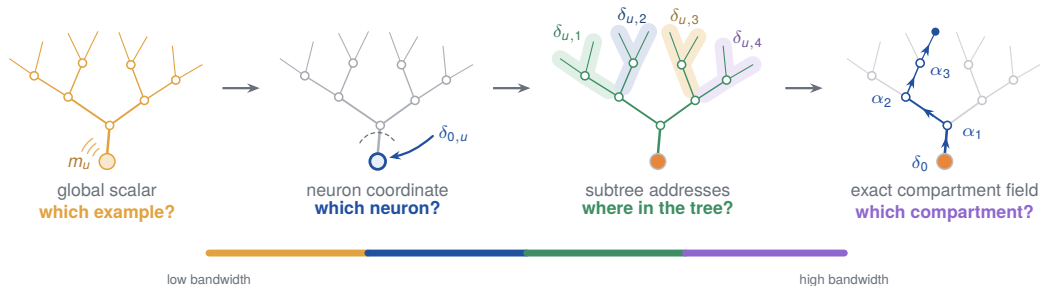
delivered along the tree

Exact reconstruction matched autodiff across 100 diagnostic runs; relative discrepancies were at numerical precision.

Local learning becomes a question of how faithfully biology approximates the transported field.

Credit fields form a hierarchy of spatial resolution

Each rung preserves more destination information and costs more feedback bandwidth.



Route gain rescales an available coordinate; it does not add feedback bandwidth.

Neuron identity and within-tree address are distinct credit-assignment problems.

A credit operator predicts when restricted routes help

teaching field $\hat{\mathbf{g}} = \mathbf{g} + \boldsymbol{\xi}$, update $\mathbf{w}^+ = \mathbf{w} - \eta M \hat{\mathbf{g}}$, noise covariance Σ

DESCENT BOUND

$$\mathbb{E}\mathcal{L}(\mathbf{w}^+) \leq \mathcal{L}(\mathbf{w}) - \eta \underbrace{\mathbf{g}^T M \mathbf{g}}_{\text{retained task signal}} + \frac{L\eta^2}{2} \left(\underbrace{\|M\mathbf{g}\|_2^2}_{\text{curvature cost}} + \underbrace{\text{tr}(M\Sigma M^T)}_{\text{admitted noise}} \right)$$

OPTIMIZED ONE-STEP UTILITY

$$U(M) = \frac{[\mathbf{g}^T M \mathbf{g}]^2}{2L\{\|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^T)\}}, \quad \mathbf{g}^T M \mathbf{g} > 0$$

Positivity is feedback alignment; the expression is exact for the matched quadratic and otherwise orders guarantees.

A restricted route helps only if what it preserves outweighs what it discards or admits as noise.

Alignment × bandwidth defines the phase boundary

This is the theory-guided prediction; the populated evidence map returns near the end.

UNIT-STEP IDENTITY-QUADRATIC CROSSOVER

$$\underbrace{\text{tr}[(I - P)\Sigma]}_{\text{rejected noise}} > \underbrace{\|(I - P)\mathbf{g}\|_2^2}_{\text{discarded task signal}}$$

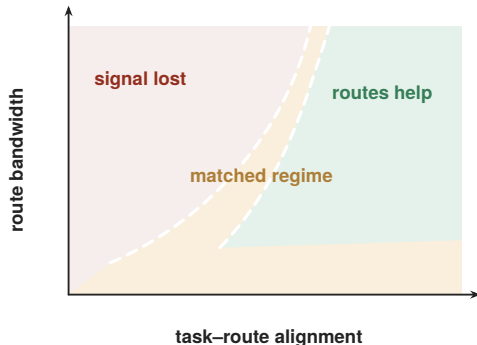
Too few routes discard signal; too many routes admit fine-scale noise. An interior bandwidth can therefore be optimal.

$$\rho_S = 0.937$$

predicted vs. observed
one-step progress

$$\rho_S = 0.916$$

predicted utility vs. final
held-out accuracy

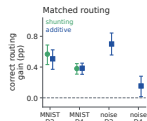
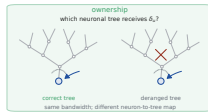
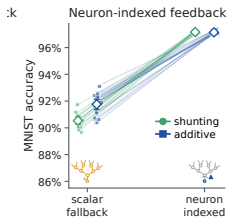


schematic prediction, not a measured phase diagram

The theory predicts both the wins and the nulls before anatomy enters the analysis.

Neuron identity closes most of the ordinary classification gap

Bandwidth selects the neuron; ownership supplies a smaller map-specific refinement.



same coordinates and bandwidth; only the neuron-to-tree assignment changes

7.8–11.2 pp

strict scalar → neuron-specific feedback on MNIST

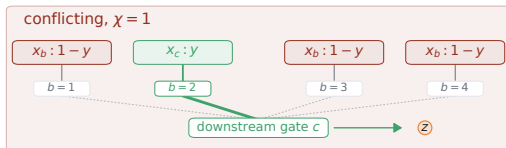
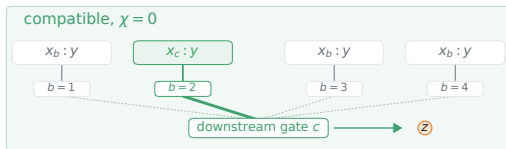
The dominant bottleneck is neuron identity; routing inside the chosen neuron is a second, smaller problem on these tasks.

+0.15 to +0.70 pp

correct ownership, favored in 58/60 paired comparisons

Credit conflict creates demand for a branch address

Every branch gets a nonzero view; context selects the task-relevant one downstream.



CONFLICT DOSE

With probability χ , every nonselected view carries the opposite class; otherwise it agrees. The selector lies downstream of branch-local eligibility, so all B local traces remain nonzero.

correct path

$$\delta_b = \delta \mathbf{1}[b = c]$$

neuron shared

$$\tilde{\delta}_b = \delta / B$$

ANALYTIC BOUNDARY

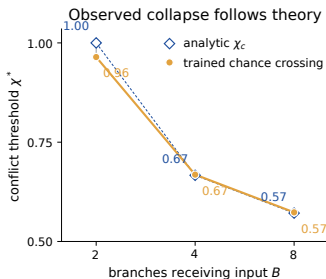
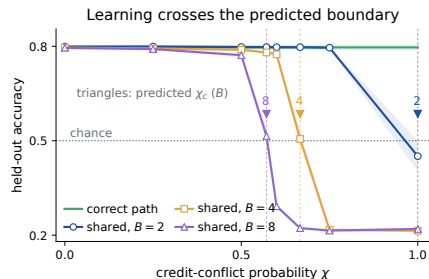
$$C_\chi = (1 - 2\chi)\mathbf{1}\mathbf{1}^T + 2\chi I_B,$$

$$\lambda_{\parallel} = B - 2\chi(B - 1), \quad \chi_c(B) = \frac{B}{2(B - 1)}.$$

This isolates a need for branch-selective information, not a dendrite-exclusive material implementation.

The path benefit appears at the predicted conflict boundary

Twenty confirmatory seeds; correct path, analytic backpropagation, and gated-point emulation coincide.



— NO CONFLICT

$\chi = 0$: mean
correct-minus-shared
difference ≤ 0.16
percentage points.

+58.2 pp

correct path minus shared
credit at $B = 4$, $\chi = 1$

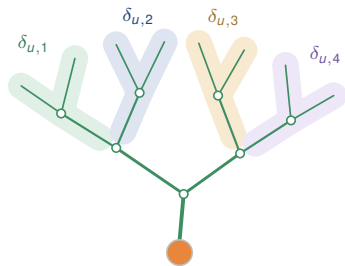
— ORDERED TRANSITION

$\chi_8^* < \chi_4^* < \chi_2^*$
utility and trained accuracy;
20/20 seeds

Branch-selective credit pays only when nonselected branches carry conflicting evidence.

Ancestry routes help only at intermediate bandwidth

The benefit is consistent with an interior-bandwidth regime and disappears at full rank.

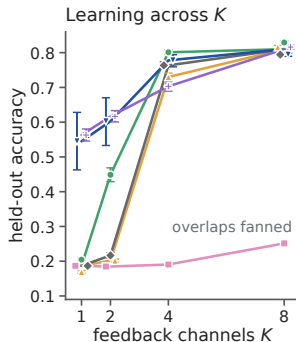


four nested ancestry routes

+1.27 pp at $K = 4$

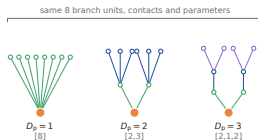
versus the best rank-matched non-anatomical

Anatomical ancestry is conditionally useful, not universally optimal.



Physical depth helps only for task-matched serial computation

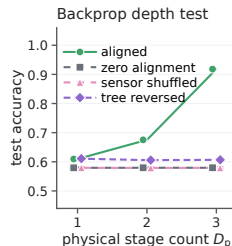
Depth is an inductive bias, not a free source of accuracy.



$$V_n = \frac{E_n + C_n}{1 + E_n + I_n + G_n + \epsilon}$$

+31 pp

aligned D3 minus D1 on a constructed multiplicative hierarchy, with branch units and parameters matched

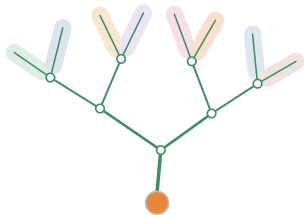


Independent sensors, shuffled sensors, and reversed order remain flat; at fixed budgets, depth per se reduced accuracy in 40/40 seeds.

Serial dendrites help only when the hierarchy of computation matches the hierarchy of the task.

Reconstructed arbors provide sparse candidate routes

Morphology supplies capacity; it does not establish that endogenous learning uses it.



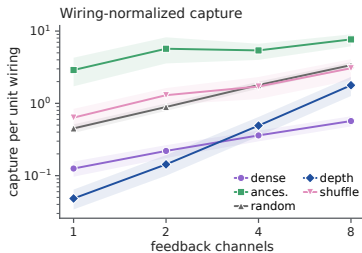
an ancestry dictionary with eight channels

85%

dense-field capture at
approximately 7% of
dense wiring

2.7×

capture per connection
beyond a
density-matched shuffle

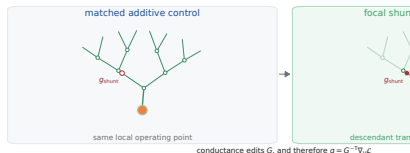


Most of the gain is sparsity; morphology adds a bounded anatomy-specific margin.

Real arbors are economical route dictionaries, but route availability is not evidence of route use.

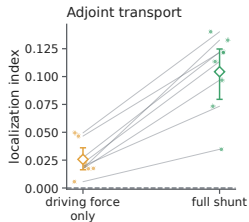
Focal shunting changes descendant transport conditionally

The established result is transport selectivity, not a generic improvement in trained learning.

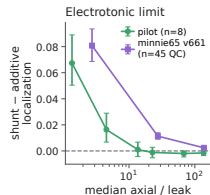


0.069

shunt-minus-additive localization contrast;
matched local voltage, somatic state, and
output error



freezing the adjoint removes the contrast;
restoring full transport recovers it

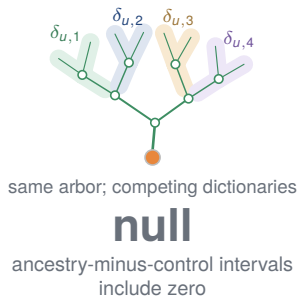


Null at textbook resting calibration; present in
high-conductance regimes. The exploratory
adaptive-shunting study did not improve final
loss over no shunt.

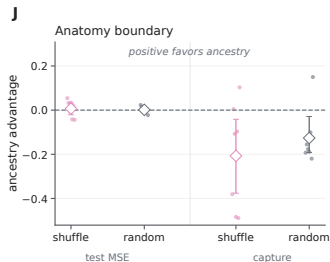
Conductance can edit route gain locally, but this does not imply an unconditional learning advantage.

Measured responses show no morphology-specific alignment

Complete-tree visual-response credit is not better captured by ancestry routes than matched controls.



Exact fields fit best; ancestry and matched low-rank controls remain statistically similar.



Measured visual responses show no endogenous alignment with ancestry routes.

Task alignment rescues the same anatomical dictionary

Anatomy and channels stay fixed; only the task-gradient direction rotates.

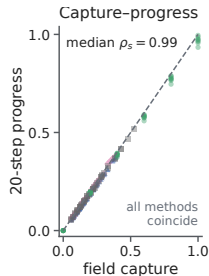
ALIGNMENT DOSE

$$\mathbf{g}(a) = \sqrt{a} \mathbf{u}_{\parallel} + \sqrt{1-a} \mathbf{u}_{\perp}, \quad \|Q Q^T \mathbf{g}(a)\|^2 = a$$

$$\rho_s = 0.99$$

field capture predicts 20-step progress

At $a = 0$, nested routes lose to matched controls in 8/8 cells; at $a = 1$, they win in 8/8.

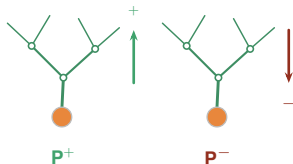


Same cells, rank, gradient energy, and curvature; only alignment changes.

Alignment can turn anatomical route capacity into a learning benefit.

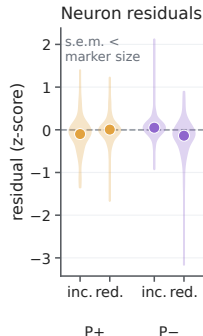
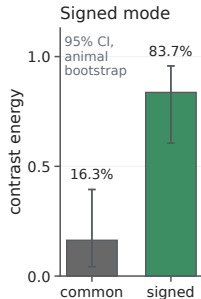
Animal learning is consistent with signed neuron coordinates

This is consistent with the first spatial rung; within-tree transport remains unmeasured.



83.7%

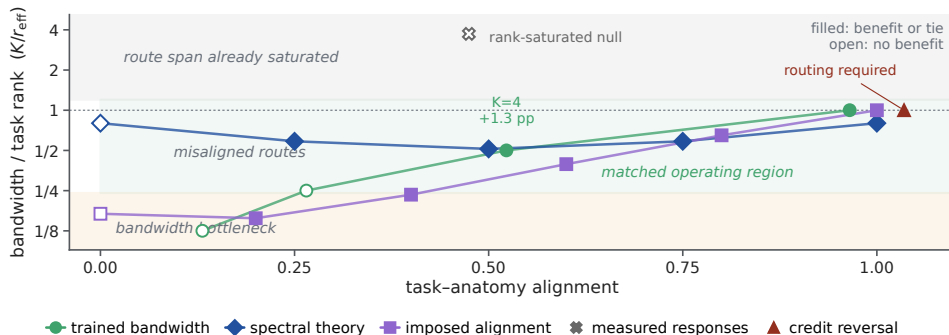
of contrast energy lies in the signed, neuron-specific mode; 6/6 animals causal signs were fixed by the BCI mapping and predict opposite dendritic contrasts



The animal result is consistent with signed neuron coordinates; routing within a tree remains open.

One phase plane organizes the positive and null results

Markers are measured or simulated outcomes; the background is the theory-guided regime map.



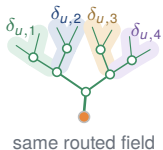
The same signal–noise theory explains where dendritic routes help and where they do not.

Dendrites change the resource ledger, not the exact ceiling

The contribution is structured local credit routing, conditional on alignment.

- EXACT CEILING

A point system given the same routed field matches the dendritic implementation. Exact backpropagation remains the first-order ceiling.



- WHAT THE TREE CONTRIBUTES

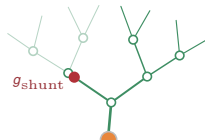
1. local conductance state
2. reusable within-neuron addresses
3. branch-local route-gain control
4. sparse feedback wiring

85% at 7%

dense capture at a small fraction of dense wiring

- DECISIVE EXPERIMENTS

- focal inhibition should shift plasticity below the branch
- the shift should scale with dose and input resistance, then saturate
- useful routing should track task-route alignment, not morphology alone

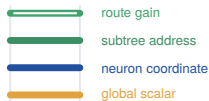


Coordinates select neurons; trees address synapses; conductance tunes route gain; alignment decides whether any of it pays.

BACKUP

Derivations, controls, audits

For questions



Appendix: the adjoint chain, from implicit function to path product

GENERAL ADJOINT

$$\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = \mathbf{0}, \quad J_V = \frac{\partial \mathbf{F}}{\partial \mathbf{V}}, \quad J_V^T \mathbf{q} = \nabla_V \mathcal{L}$$

$$\frac{\partial \mathcal{L}}{\partial g_i} = -\mathbf{q}^T \frac{\partial \mathbf{F}}{\partial g_i} = x_i(E_i - V_n) q_n$$

synapse-local factor $\times$ compartment adjoint

In a reciprocal passive cable, $J_V = G$ is symmetric positive definite and $G^{-T} = G^{-1}$: the adjoint is a Green's-function solve. For a general nonsymmetric or recurrent compartmental system the inverse transpose is essential.

This fixed-point adjoint is the steady-state member of the Almeida–Pineda recurrent-backpropagation lineage.

With the residual convention $\mathbf{F} = G\mathbf{V} - \mathbf{b}$. For the triangular tree system, $q_n = R_n^{\text{tot}} \partial \mathcal{L} / \partial V_n$, so the adjoint and compartment factorizations are the same identity with the input-resistance factor moved between terms.

Credit is always eligibility times an adjoint; only the adjoint changes.

Appendix: deriving $U(M)$ and its exactness caveat

DESCENT LEMMA

For L -Lipschitz $\nabla \mathcal{L}$ and $\mathbf{w}^+ = \mathbf{w} - \eta M \hat{\mathbf{g}}$, $\hat{\mathbf{g}} = \mathbf{g} + \xi$:

$$\begin{aligned}\mathbb{E} \mathcal{L}(\mathbf{w}^+) &\leq \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top M \mathbf{g} \\ &\quad + \frac{L\eta^2}{2} (\|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^\top))\end{aligned}$$

Optimizing the right side over η when $\mathbf{g}^\top M \mathbf{g} > 0$:

$$U(M) = \frac{[\mathbf{g}^\top M \mathbf{g}]^2}{2L \{ \|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^\top) \}}$$

CAVEAT

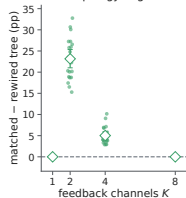
The bound is an upper guarantee for a general L -smooth loss and is exact for the corresponding quadratic when curvature equals L on the update directions; comparisons of $U(M)$ otherwise order guarantees, not necessarily realized losses.

Diagnostics: $U(M)$ predicted norm-matched one-step progress with Spearman $\rho = 0.937$ and final held-out accuracy with $\rho = 0.916$ across 540 dendritic-route conditions; spectral capture alone predicted one-step progress at $\rho = 0.877$.

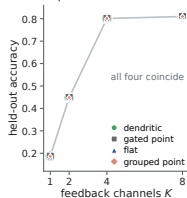
One-step geometry predicts within-family progress; trajectory-accrued advantages can escape it.

Appendix: factorial control battery and implementation equivalence

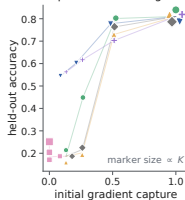
Task-topology alignment



Representation match



Capture and learning



■ TOPOLOGY ALIGNMENT

matched tree minus rewired tree: +0.231 at $K=2$, +0.050 at $K=4$; 20/20 paired seeds each.

■ REPRESENTATION MATCH

dendritic, gated-point, flat and grouped-point implementations coincide exactly (maximum accuracy difference 0) when given identical routed fields.

■ CAPTURE IS NOT FATE

fields with similar initial capture but different route signs and task alignment learned differently.

An address resource any point system can buy with the same routes.

Appendix: input-validity audit and clean exact/BP agreement

OUTCOME-INDEPENDENT RULE

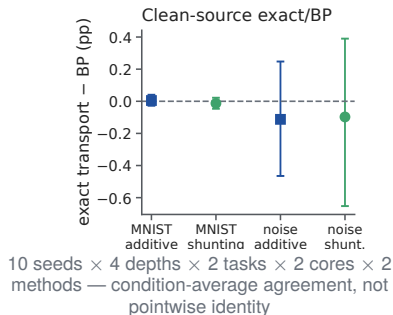
signed synthetic drive is invalid for a positive-conductance shunting denominator: 640 signed-shunting runs are invalid, and the 400-run inhibitory-dose family that depends on them is excluded with its 200 valid partners — 840 of 1,840 runs leave inference; all remain in the ledger.

CLEAN-SOURCE AUDIT

all 320 detached clean-source runs passed the configuration, log, metric and checkpoint audit; synthetic shunting uses a non-negative transfer.

AGREEMENT

all four depth-averaged exact-minus-BP intervals include zero; mean across 160 pairs -0.055 pp; largest single pair difference 5.90 pp.



The clean audit supports agreement on average, never formal equivalence.

Appendix: reliability gain, and same-span conditioning

FINITE-STEP GAIN

For disjoint branches with signal S_b , noise N_b , step $\eta = c/L$:

$$\mathcal{G}_c(\mathbf{a}) = \frac{c}{L} \sum_b \left[a_b S_b - \frac{c a_b^2}{2} (S_b + N_b) \right]$$

$$a_b^* = \min \left\{ 1, \frac{S_b}{c(S_b + N_b)} \right\}$$

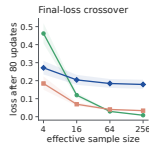
A point model supplied with identical branch gates must match — and did, to numerical precision.

trained branch model, 50 fresh seeds: aligned shunting beat the best global gain by 0.001817 in one-step loss reduction (49/50); final aligned-minus-no-shunt loss was null (0.000425, $P = 0.154$)

SAME-SPAN FIELD ITERATION

$$\mathbf{f}_{t+1} = \mathbf{f}_t - c\mathbf{A}(\mathbf{f}_t - \mathbf{g} - \boldsymbol{\xi}_t)$$

$$\mathbb{E} \mathcal{L}_T = \frac{1}{2} \sum_j \left[B_{j,T} + \frac{\sigma^2}{n} V_{j,T} \right]$$

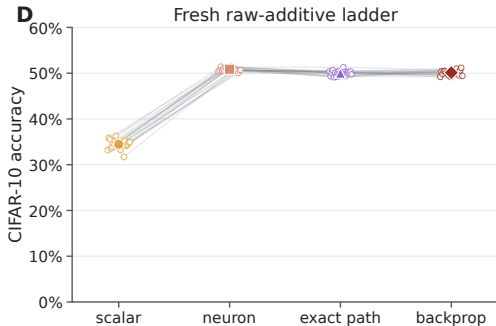


rank-eight same-span coordinates, condition numbers 1 / 15 / 388.5: slow modes filter noise at small n , orthogonal modes cut bias at large n ; predicted crossovers $n^* = 38.5, 8.25$

Conductance realizes the predicted ordering; conditioning trades bias against variance.

Appendix: harder data isolates the feedback bottleneck

Fresh 20-seed raw-additive confirmation; flattened CIFAR-10 is a transfer assay, not a vision benchmark.



Gray lines pair seeds; large symbols show means and 95% intervals.

Harder data magnifies neuron identity, not a generic benefit of finer dendritic transport.

■ ADEQUATE CONTROL

matched BP: 50.12%; all 80 fits passed source, calibration, checkpoint and convergence audits.

■ NEURON IDENTITY

+16.39 pp over strict scalar (95% CI 15.88–16.91; 20/20 seeds).

■ PATH BOUNDARY

exact path was 0.86 pp below neuron-specific feedback; the prespecified path-resolution gate failed.

■ IMPLEMENTATION CHECK

exact path was equivalent to matched BP within ± 1 pp (TOST $P = 1.58 \times 10^{-5}$).

Appendix: references

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All quantitative claims trace to the journal manuscript and its source-data package.